

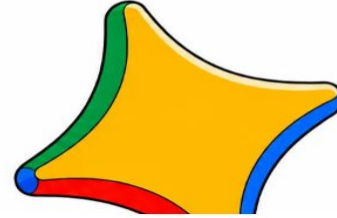


Build with AI

Code for Communities

2nd Edition

India's developers. World's challenges.



Team Details

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- d. Problem Statement: AI for Digital Public Infrastructure & Governance



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Brief about your solution and problem statement addressing

Governments receive development-related requests through websites, WhatsApp, SMS, voice calls, call centres, and public forums. However, this information is often fragmented across departments, languages, and systems. Similar requests may be treated as separate complaints, while communities with limited connectivity or low digital participation may not be visible at all. This makes it difficult for policymakers to identify the most urgent and widespread development needs.

We are creating an AI-assisted Public Development Intelligence Platform that works behind existing citizen communication channels. It connects with government websites through APIs, receives WhatsApp and SMS messages through webhooks, and processes voice-based inputs through speech-enabled integrations. Citizens can continue using familiar channels without downloading a new application.

The platform will:

- Collect citizen complaints, requests, observations, proposals, follow-ups, and supporting evidence.
- Use Google AI and Gemini to understand multilingual text and voice submissions.
- Extract the issue, location, affected service, urgency, and citizen intent.
- Group similar submissions across different languages, channels, and wording.
- Combine citizen signals with demographic, infrastructure, and public-investment data.
- Identify demand hotspots where multiple forms of evidence indicate unmet need.
- Highlight civic data blind spots where low participation may hide serious problems.
- Present priorities using four transparent dimensions: need, confidence, equity, and actionability.
- Provide policymakers with evidence-backed recommendations for investigation and intervention.
- Record official decisions, actions, and progress for accountability.

The platform is not an autonomous government decision-maker. It does not approve projects or allocate public funds. Instead, it provides policymakers with a unified, contextual, and explainable view of public need, helping them make more inclusive and evidence-based development decisions.

1. How different is it from any of the other existing ideas?

Most existing government systems focus on collecting and tracking individual complaints. Our platform focuses on understanding broader development needs by connecting multiple citizen signals and public data sources.

Our key differences are:

- Works with existing channels: Integrates with websites, WhatsApp, SMS, voice systems, call centres, and government portals instead of forcing citizens to adopt a new application.
- Understands multilingual inputs: Uses Google AI and Gemini to process different Indian languages, mixed-language messages, voice inputs, and informal descriptions.
- Analyses patterns, not just complaints: Groups similar requests across channels and languages to identify systemic issues.
- Adds public context: Combines citizen feedback with demographic, infrastructure, and public-investment data.
- Identifies both hotspots and blind spots: Finds areas with high visible demand as well as communities where low participation may hide serious needs.
- Provides explainable priorities: Separates need, confidence, equity, and actionability instead of producing one unexplained score.
- Keeps humans accountable: Supports policymakers with evidence and recommendations, while final decisions remain with authorised officials.
- Creates an audit trail: Records what was reviewed, what decision was made, and what action followed.

2. How will it be able to solve the problem?

The platform converts fragmented citizen communication into structured and comparable development intelligence.

It will:

1. Collect citizen signals from existing communication channels.
2. Use Google AI to understand the language, meaning, issue, location, and urgency.
3. Detect related submissions and group them into common development needs.
4. Compare these signals with demographic, infrastructure, and investment information.
5. Identify demand hotspots, underserved areas, and data blind spots.
6. Rank issues using transparent criteria such as need, confidence, equity, and actionability.
7. Present policymakers with evidence-backed investigation and intervention options.
8. Capture official decisions and progress, creating a feedback loop between citizens and government.

List of features offered by the solution

- **Omnichannel Citizen Input**

Collects requests through websites, WhatsApp, SMS, voice calls, call centres, and public forums.

- **Multilingual AI Understanding**

Uses Google AI and Gemini to understand Indian languages, mixed-language text, informal descriptions, and voice submissions.

- **Structured Signal Extraction**

Identifies the issue, location, affected service, urgency, intent, and supporting evidence.

- **Cross-Channel Issue Clustering**

Groups similar requests even when they come from different channels, languages, or wording.

- **Public Data Enrichment**

Combines citizen signals with demographic, infrastructure, historical, and public-investment data.

- **Demand Hotspot Detection**

Identifies locations where multiple signals indicate a concentrated or repeated development need.

- **Civic Data Blind Spot Detection**

Highlights areas where low participation may hide infrastructure gaps or underserved communities.

- **Interactive Geographic View**

Displays development needs, hotspots, infrastructure gaps, and priority areas on maps.

- **Explainable Prioritisation**

Evaluates each issue across four dimensions: need, confidence, equity, and actionability.

- **Evidence-Based Recommendations**

Suggests issues for investigation and possible intervention without making automatic government decisions.

- **Policymaker Dashboard**

Provides officials with summaries, evidence, public context, priority rankings, and uncertainty indicators.

- **Human Review and Decision Control**

Allows authorised officials to accept, reject, investigate, or request additional evidence.

- **Audit and Accountability Trail**

Records decisions, rationales, actions, status updates, and outcomes.

- **Citizen Feedback Loop**

Enables citizens to receive updates and contribute follow-up information about reported issues.

- **Privacy and Access Controls**

Protects sensitive information through role-based access, data minimisation, and controlled visibility.



Google Technologies used in the solution

Google technologies are central to both the intelligence and deployment of the platform. Google AI is used to understand citizen inputs, while Google Cloud provides the infrastructure required to process, store, secure, and visualise the information.

Google AI and Intelligence Layer

- **Gemini through Vertex AI**
Understands multilingual text, mixed-language messages, informal descriptions, and voice-based submissions.
- **Gemini Structured Output**
Converts unstructured citizen inputs into structured fields such as issue, location, intent, urgency, and affected service.
- **Gemini Embeddings**
Helps identify similar requests across different languages, channels, and wording.
- **Gemini Summarisation and Explanation**
Creates concise summaries of issue clusters and explains why a particular development need requires attention.
- **Vertex AI Model Evaluation**
Supports testing and evaluation of AI outputs for accuracy, consistency, language coverage, and responsible use.

Role of Google AI

Google AI does not make final government decisions. It helps the platform understand, organise, compare, summarise, and explain citizen evidence. Policymakers remain responsible for reviewing the evidence and deciding what action should follow.

Technology availability and exact service selection should be validated again during implementation.

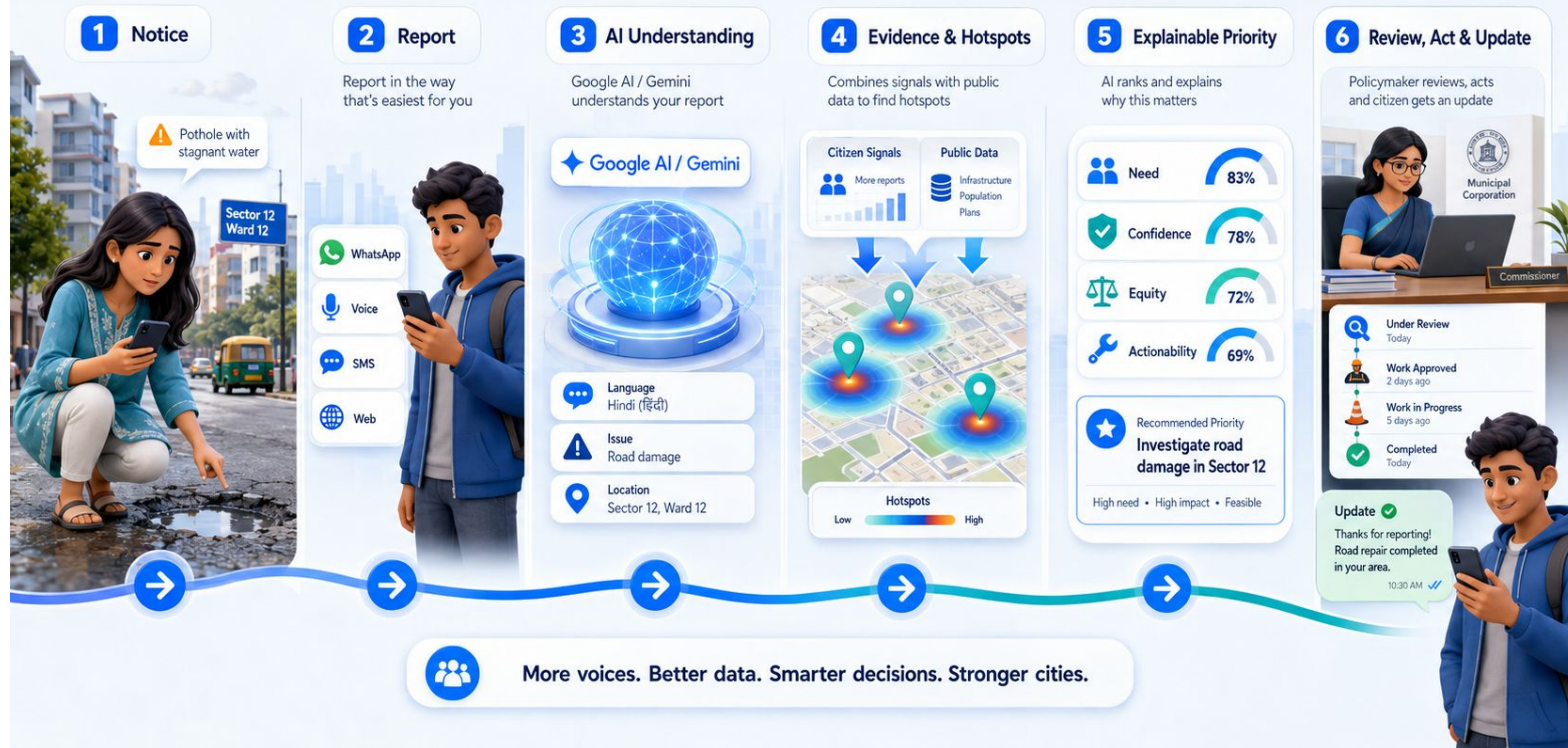
Google Cloud Platform

- **Cloud Run**
Hosts the API layer, processing services, AI workflows, and policymaker dashboard as scalable containerised services.
- **Cloud Pub/Sub**
Acts as the asynchronous messaging layer between citizen input channels and processing services, allowing the system to handle multiple submissions reliably.
- **Cloud SQL for PostgreSQL**
Stores structured citizen signals, public datasets, issue clusters, priority scores, official decisions, and audit records.
- **Google Maps Platform**
Supports geocoding, location normalisation, geographic visualisation, and demand-hotspot mapping.
- **Firebase Authentication**
Provides secure authentication for policymakers, administrators, analysts, and other authorised users.



Process flow diagram or Use-case diagram

From Citizen Voice to Public Action

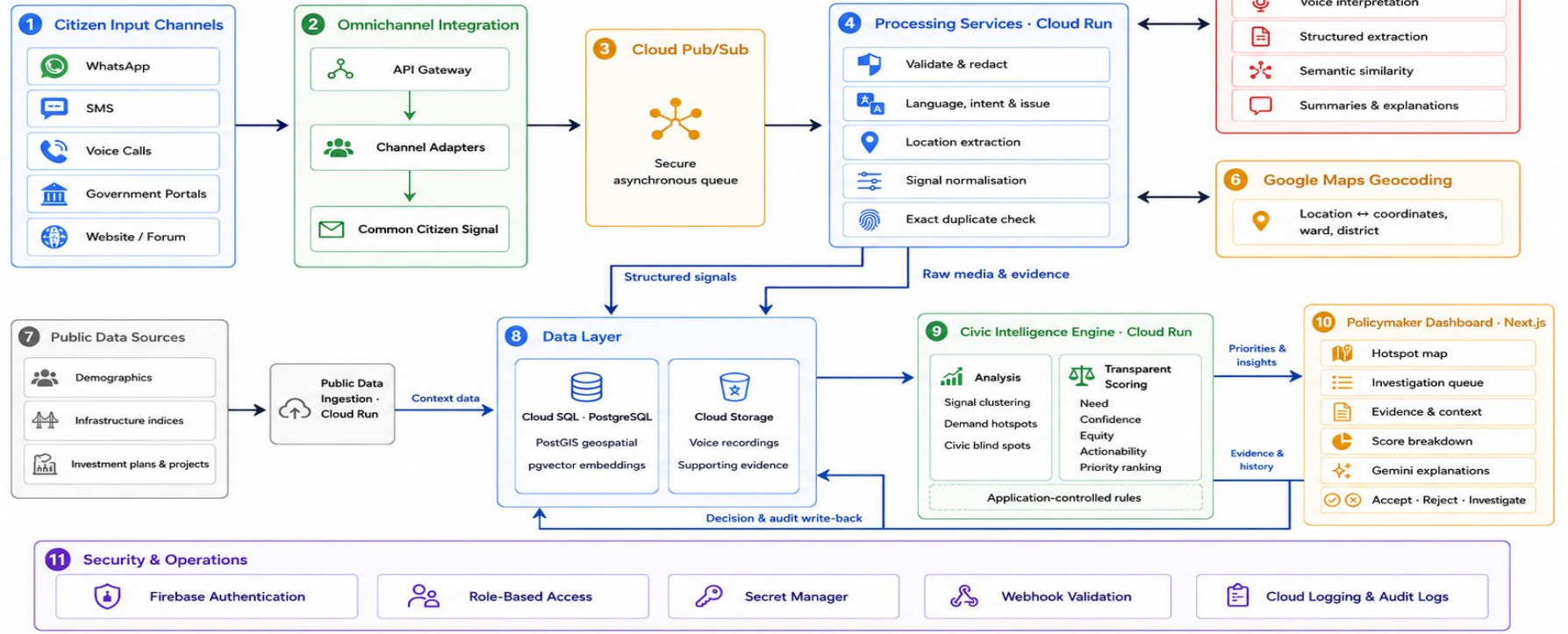




Architecture diagram of the proposed solution

AI-Assisted Public Development Intelligence Platform

Omnichannel citizen signals → Google AI enrichment → transparent civic prioritisation



Why Our Solution Is Deployable

Our platform is designed to integrate with the systems that governments already use, making adoption practical and incremental. It does not require citizens to download a new application or change their existing behaviour.

- **Works with existing channels:** Connects with government websites, WhatsApp, SMS, voice-call systems, call centres, and public forums through APIs, webhooks, and integration adapters.
- **Can start with a focused pilot:** The solution can first be deployed for one city, district, department, or development issue such as roads, water, healthcare, or sanitation.
- **Built on scalable Google Cloud services:** Cloud Run, Pub/Sub, Cloud SQL, Cloud Storage, Vertex AI, and Google Maps Platform support modular processing, storage, geolocation, and scaling.
- **Modular architecture:** New channels, languages, datasets, departments, and workflows can be added without rebuilding the entire platform.
- **Human-in-the-loop governance:** The platform provides recommendations and evidence, while authorised officials retain control over investigation, approval, budgeting, and implementation.
- **Explainable and auditable:** Each recommendation can show its source signals, public-data context, confidence level, prioritisation factors, uncertainty, and official decision history.
- **Responsible data handling:** Access controls, data minimisation, secure storage, and role-based permissions help protect sensitive citizen information.
- **Designed as reusable public infrastructure:** The common data model and integration layer can be adapted by different government departments, states, and countries.



Scalability Across India and BRICS Countries

The platform is designed as a modular Digital Public Good that can begin with a focused pilot and expand across different regions, government departments, languages, and administrative systems.

- **Pilot-to-national expansion:** Start with one city, district, or department and gradually expand across states and national programmes.
- **Multilingual by design:** Add Indian languages, regional dialects, mixed-language inputs, and local terminology without changing the core platform.
- **Modular integration layer:** Connect with different government websites, portals, WhatsApp systems, SMS providers, call centres, and departmental applications through APIs and adapters.
- **Flexible public-data model:** Adapt to different demographic datasets, infrastructure indicators, investment plans, administrative boundaries, and policy priorities.
- **Cloud-based scalability:** Google Cloud services can support increasing volumes of citizen inputs, AI processing, storage, and policymaker users.
- **Department-specific use cases:** Extend from roads, water, healthcare, and sanitation to education, transport, housing, public safety, and other development areas.
- **Low-connectivity inclusion:** Support voice, SMS, assisted reporting, and local outreach so that participation is not limited to smartphone or internet users.
- **BRICS adaptability:** The same core model can be localised for Brazil, Russia, India, China, and South Africa by adapting languages, government structures, privacy laws, data standards, communication channels, and local development priorities.
- **Reusable public infrastructure:** The platform's common signal format, integration approach, explainability principles, and governance model can be reused across jurisdictions.



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Expected Impact and Outcomes

The platform will help governments move from fragmented and reactive complaint handling to proactive, inclusive, and evidence-based development planning.

- Faster identification of public needs: Recognises repeated and systemic issues across multiple channels.
- More inclusive participation: Gives voice to citizens who may prefer regional languages, voice calls, SMS, or assisted reporting.
- Better visibility for underserved communities: Highlights civic data blind spots where low participation may hide serious infrastructure or service gaps.
- Improved development prioritisation: Helps officials compare need, confidence, equity, and actionability before deciding what requires investigation.
- Better use of public resources: Connects citizen needs with existing infrastructure and public-investment plans, reducing duplication and improving alignment.
- More transparent decisions: Shows the evidence and reasoning behind each recommended priority.
- Stronger accountability: Records official reviews, decisions, actions, and progress from the original citizen signal to the final outcome.
- Improved coordination: Provides departments and planning teams with a common view of citizen needs and development gaps.
- Continuous feedback loop: Allows citizen follow-ups and implementation updates to improve future analysis.

Long-Term Outcome

More visible needs, smarter public investment, stronger citizen trust, and more equitable development decisions



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Additional Details/Future Development

The proposed solution is designed as a modular and scalable public digital infrastructure layer. The initial version will demonstrate multilingual citizen input, Google AI-based understanding, issue clustering, public-data enrichment, hotspot detection, explainable prioritisation, and policymaker review. Future development will expand the platform's coverage, intelligence, integrations, and accountability capabilities.

Future Development Areas

- **More communication channels**
Add additional messaging platforms, regional call centres, mobile applications, government portals, and offline data-entry systems.
- **Expanded language coverage**
Support more Indian languages, dialects, mixed-language inputs, and region-specific terminology.
- **Richer evidence processing**
Analyse images, documents, videos, location data, and field-verification reports along with text and voice submissions.
- **Advanced public-data integration**
Connect with live infrastructure registers, census information, service availability data, project-monitoring systems, and public-investment databases.
- **Proactive community outreach**
Identify civic data blind spots and support targeted surveys, field visits, assisted reporting, and local consultations.
- **Department-specific workflows**
Create customised workflows for roads, water, healthcare, sanitation, transport, education, and other development areas.
- **Project and outcome monitoring**
Track whether approved interventions were implemented and whether citizens experienced measurable improvement.
- **Public transparency features**
Provide suitable aggregate dashboards showing major issues, actions taken, and progress without exposing personal information.
- **Cross-border adaptation**
Localise the platform for other Indian states and BRICS countries by adapting languages, administrative structures, datasets, privacy requirements, and policy rules.